Establish the force model

Original slide / English

DAMPED OSCILLATOR / ORIGINAL DEMO

Forces and free motion

Measure $x(t)$ from the equilibrium position.

Spring force

$F_s = -kx$

Damping force

$F_d = -bx'$

Newton's second law connects both forces.

$mx'' = -bx' - kx$

$mx'' + bx' + kx = 0$

$m > 0$, $k > 0$, $b > 0$; constant coefficients; no driving force.

Next: which function satisfies inertia, damping and restoring force?

01 / 03

Translated slide / Chinese

阻尼振动 / 原创演示

力与自由运动

$x(t)$ 为相对平衡位置的位移。

弹簧力

$F_s = -kx$

阻尼力

$F_d = -bx'$

牛顿第二定律把两种力串联起来。

$mx'' = -bx' - kx$

$mx'' + bx' + kx = 0$

$m > 0$、$k > 0$、$b > 0$；系数恒定，无外部驱动力。

接下来：什么函数同时满足惯性、阻尼和回复作用?

01 / 03

Vocabulary

Equilibrium / 平衡位置	$x = 0$ is the coordinate origin.
Viscous damping / 黏性阻尼	A model force proportional to velocity.

Foundation bridge: Guide page 1 explains force signs, derivative notation, units and model conditions before solving. (Tutor foundation supplement; authored source slide 1)

Original practice P1: Complete original P1(a)–(d): guide page 8. All worked answers: pages 8–10. (Tutor-authored original practice; new parameters)

Source: source-slides.pdf, page 1

Study guide: pages 1–3, 8

Derive and interpret the roots

Original slide / English

DAMPED OSCILLATOR / ORIGINAL DEMO

Why oscillations decay

Try an exponential to turn differentiation into algebra.

$$x(t) = Ce^{\lambda t}$$

$$m\lambda^2 + b\lambda + k = 0$$

$$\gamma = \frac{b}{2m}, \qquad \omega_0 = \sqrt{\frac{k}{m}}$$

Underdamped condition: $0 < \gamma < \omega_0$.

$$\lambda = -\gamma \pm i\omega_{\mathrm{d}}, \qquad \omega_{\mathrm{d}} = \sqrt{\omega_0^2 - \gamma^2}$$

Real part: decay. Imaginary part: oscillation.

Next: how do initial displacement and velocity select the motion?

02 / 03

Translated slide / Chinese

阻尼振动 / 原创演示

振动为何衰减

试取指数函数，把微分关系变为代数关系。

$$x(t) = Ce^{\lambda t}$$

$$m\lambda^2 + b\lambda + k = 0$$

$$\gamma = \frac{b}{2m}, \qquad \omega_0 = \sqrt{\frac{k}{m}}$$

欠阻尼条件：$0 < \gamma < \omega_0$。

$$\lambda = -\gamma \pm i\omega_{\mathrm{d}}, \qquad \omega_{\mathrm{d}} = \sqrt{\omega_0^2 - \gamma^2}$$

实部决定衰减；虚部决定振动。

接下来：初始位移和速度如何选出具体运动?

02 / 03

Vocabulary

Characteristic root / 特征根	A rate preserved by differentiation.
Underdamped / 欠阻尼	$0 < \gamma < \omega_0$; decaying oscillation.

Connect the steps: Guide page 3 solves and interprets the roots; page 4 converts complex modes to real motion. Page 11 checks limiting regimes. (Tutor derivation; authored source slide 2)

Original practice P1(a): Complete original P1(a)–(d): guide page 8. All worked answers: pages 8–10. (Tutor-authored original practice)

Source: source-slides.pdf, page 2

Study guide: pages 2–4, 8, 10–11

Fit initial data and read the response

Original slide / English

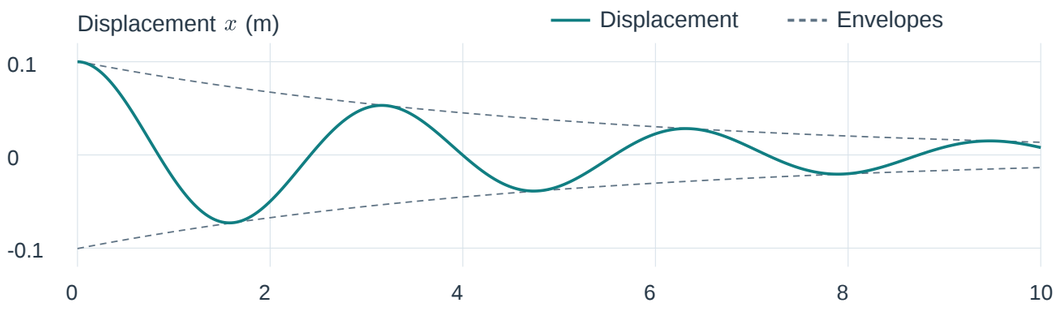
DAMPED OSCILLATOR / ORIGINAL DEMO

Initial data choose the motion

$$x(t) = e^{-\gamma t}[A\cos(\omega_d t) + B\sin(\omega_d t)]$$

$$A = x_0, \qquad B = \frac{v_0 + \gamma x_0}{\omega_d}$$

Example: $m = 1$ kg, $b = 0.4$ kg/s, $k = 4$ N/m, $x_0 = 0.1$ m, $v_0 = 0$.



Envelopes: $\pm Re^{-\gamma t}$, $R = \sqrt{A^2 + B^2}$. Displacement still oscillates.

03 / 03

Translated slide / Chinese

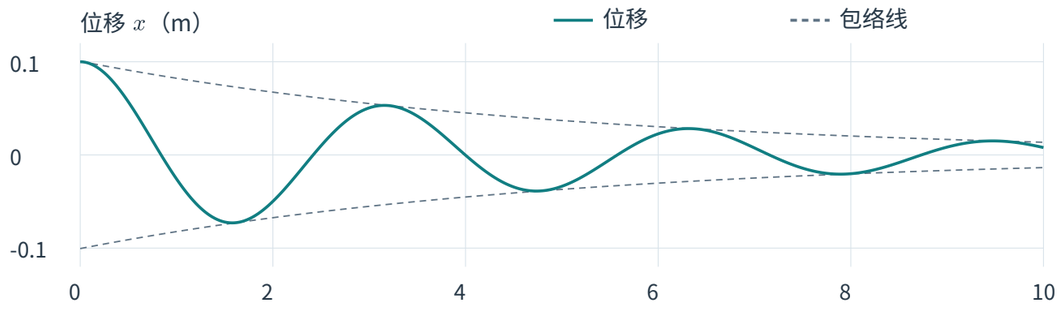
阻尼振动 / 原创演示

初值确定运动

$$x(t) = e^{-\gamma t}[A\cos(\omega_d t) + B\sin(\omega_d t)]$$

$$A = x_0, \qquad B = \frac{v_0 + \gamma x_0}{\omega_d}$$

示例：$m = 1$ kg，$b = 0.4$ kg/s，$k = 4$ N/m，$x_0 = 0.1$ m，$v_0 = 0$。



包络线为 $\pm Re^{-\gamma t}$，$R = \sqrt{A^2 + B^2}$；位移仍会振荡。

03 / 03

Vocabulary

Initial condition / 初始条件	x_0 and v_0 select one motion.
Envelope / 包络线	A bound $\pm Re^{-\gamma t}$, with $R = \sqrt{A^2 + B^2}$.

Resolve a common confusion: Guide pages 4–5 use the product rule and initial data and explain why R need not equal the initial displacement. Pages 6–7 connect the response graph to energy loss. (Tutor explanation; authored source slide 3)

Original practice P1(b–d): Complete original P1(a)–(d): guide page 8. All worked answers: pages 8–10. (Tutor-authored original practice; baseline resets explicitly)

Source: source-slides.pdf, page 3

Study guide: pages 4–7, 9–10